

PT100 Calibration Fit Report

Input source: calibration_2026-04-22

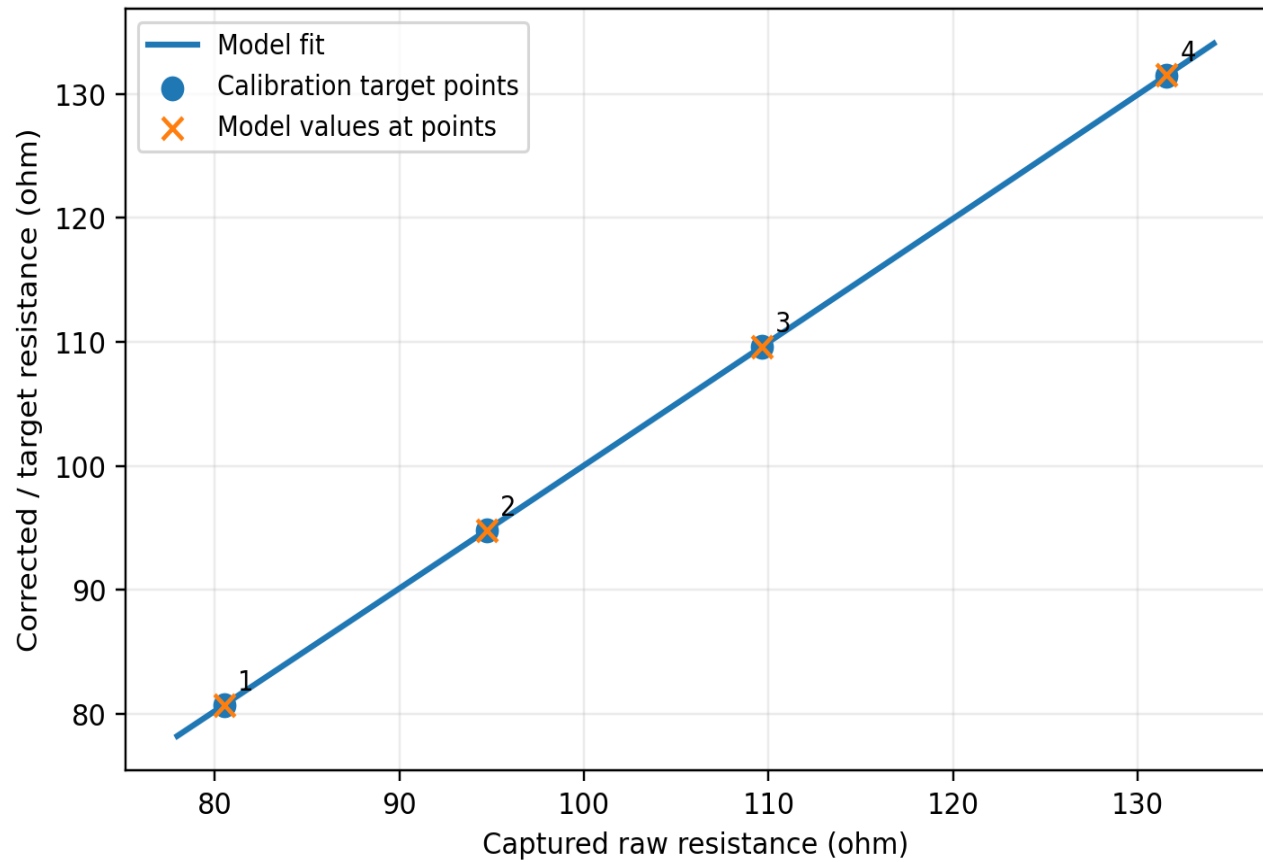
Summary

Item	Value
Input source	calibration_2026-04-22
Calibration equation	ax^2+bx+c
Fit domain	Resistance (ohms)
Model type	poly
Polynomial degree	2
Calibration points used	4
Model parameters	3
Degrees of freedom	1
Calibration method	Reference transfer from Fluke 51 II N1052
Effective last UTC	2026-04-17T00:00:00Z
Due date	2027-04-17T00:00:00Z

Calibration Curve

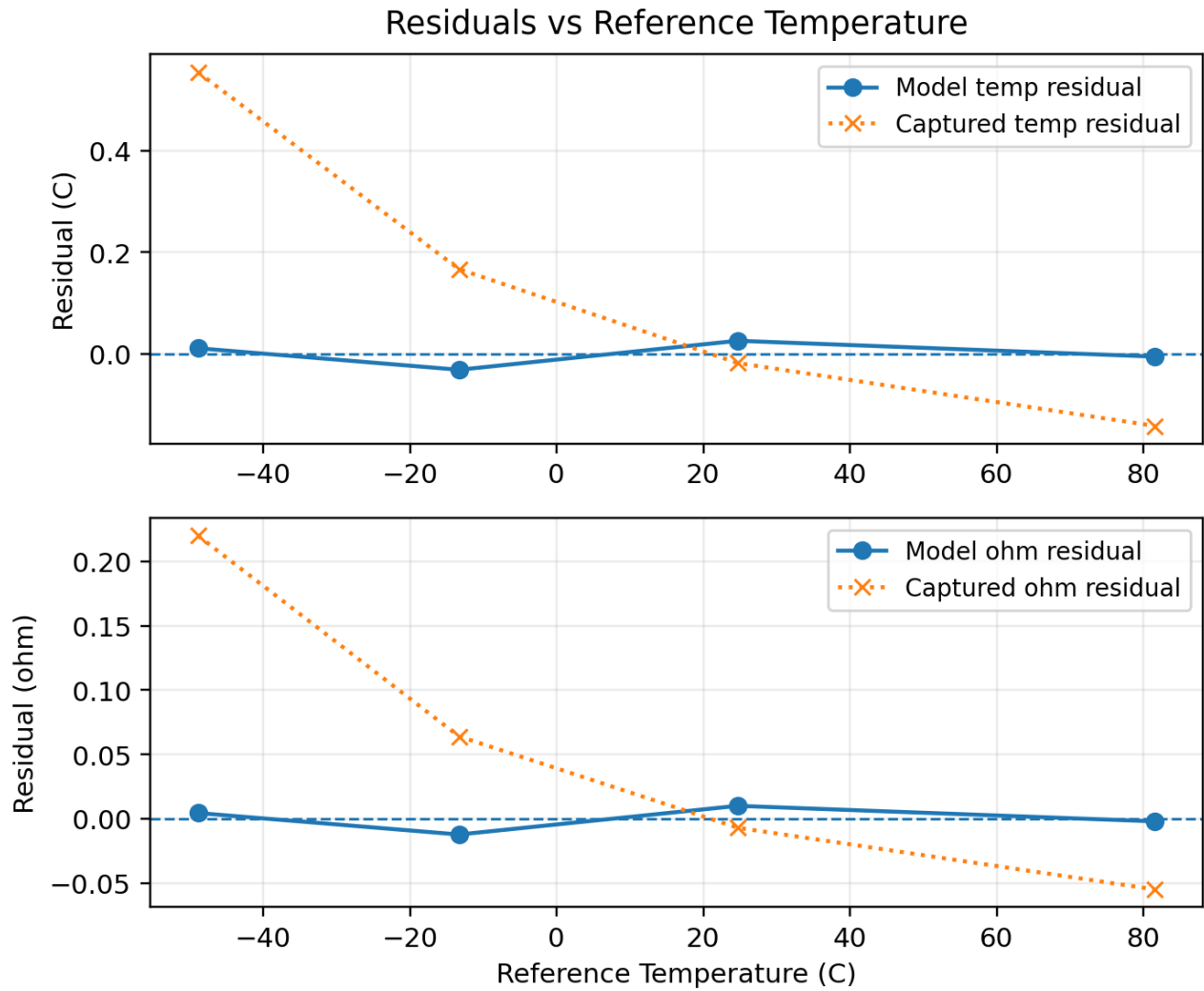
Polynomial fit with calibration target points and model values at each point.

Calibration curve - target points and fitted model



Residuals

Model-fit and captured residuals plotted against reference temperature.



Model Fit Statistics

Metric	Value
Mean absolute residual (ohms)	0.007189709
Root mean square error (ohms)	0.008283515
Maximum absolute residual (ohms)	0.012259575
Mean absolute residual (C)	0.018400459
Root mean square error (C)	0.021190599
Maximum absolute residual (C)	0.031243308
R ²	0.999999808
Adjusted R ²	0.999999424
Largest applied correction (ohms)	0.215606214
Mean signed residual (ohms)	3.40519205e-05
Residual standard deviation (ohms)	0.008283445
Sum of squared error (ohm ²)	0.000274466488
Mean signed residual (C)	8.90917565e-05
Residual standard deviation (C)	0.021190412
Sum of squared error (C ²)	0.001796166

Calibration Point Values

Pt	Target C	Target ohm	Captured C	Captured ohm	Model C	Model ohm
1	-48.900	80.743	-49.455	80.523	-48.911	80.739
2	-13.300	94.792	-13.467	94.728	-13.269	94.804
3	24.700	109.618	24.718	109.625	24.674	109.608
4	81.600	131.507	81.742	131.562	81.605	131.509

Residuals and Applied Corrections

Pt	Model res C	Model res ohm	Capture res C	Capture res ohm	Applied correction (ohm)
1	0.011163	0.004431	0.555000	0.220000	0.215606
2	-0.031243	-0.012260	0.167000	0.064000	0.075894
3	0.025816	0.010016	-0.018000	-0.007000	-0.016748
4	-0.005379	-0.002052	-0.142000	-0.055000	-0.052752